

Prototyping

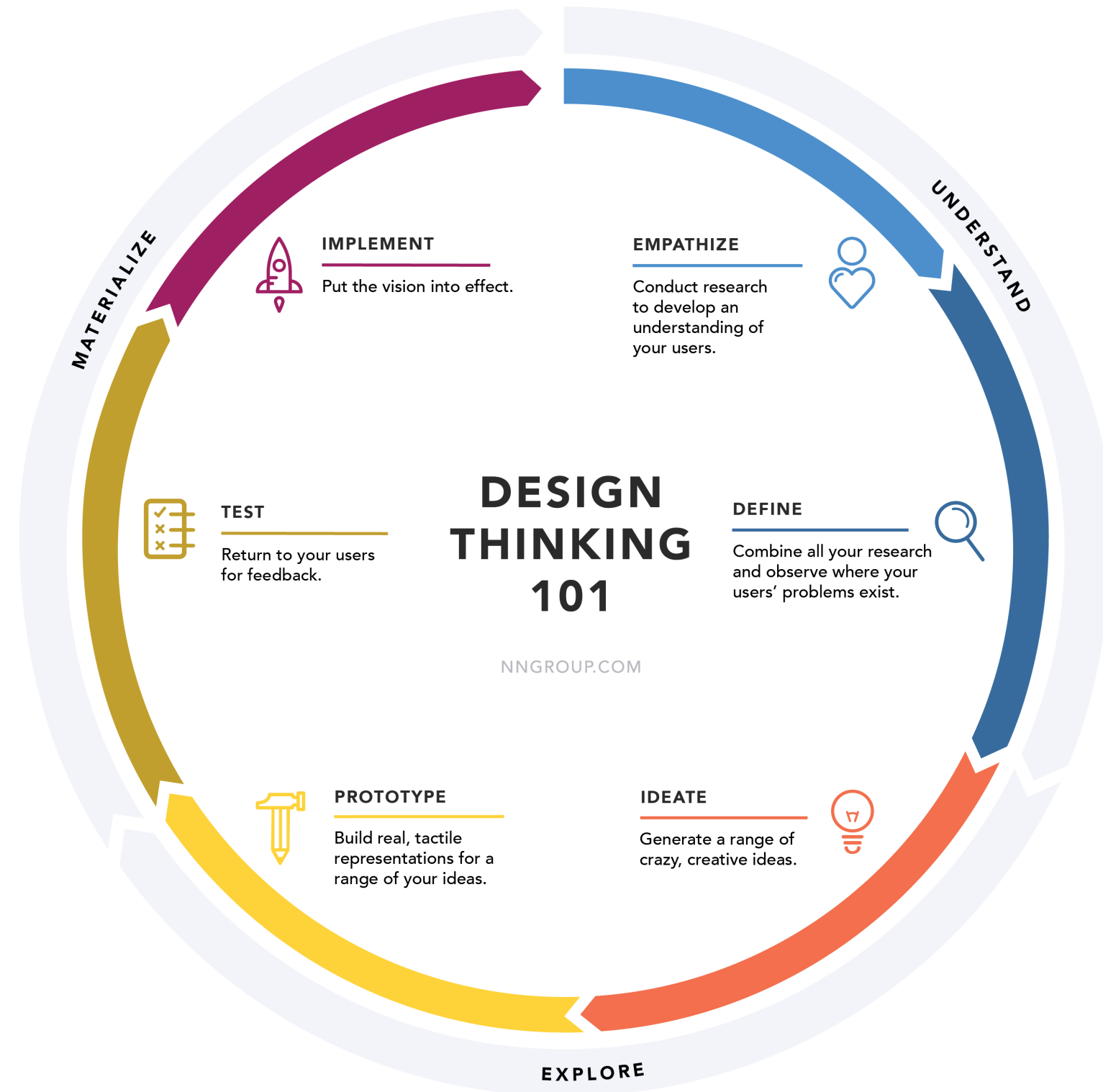
Building User Interfaces

Cole Nelson

Make a copy of today's ICA and share it with your group members!

Learning Objectives

- Understand prototyping methods and theories.
- Be able to choose an appropriate method based on the needs of the use case.
- Be able to create both *low fidelity* and *high fidelity* prototypes and identify the appropriate use for each.



Prototyping¹

Definition: Building a draft or an early version of a product or system in order to explore, demonstrate, and test design ideas for a part or the entirety of the product or system.

¹Image source



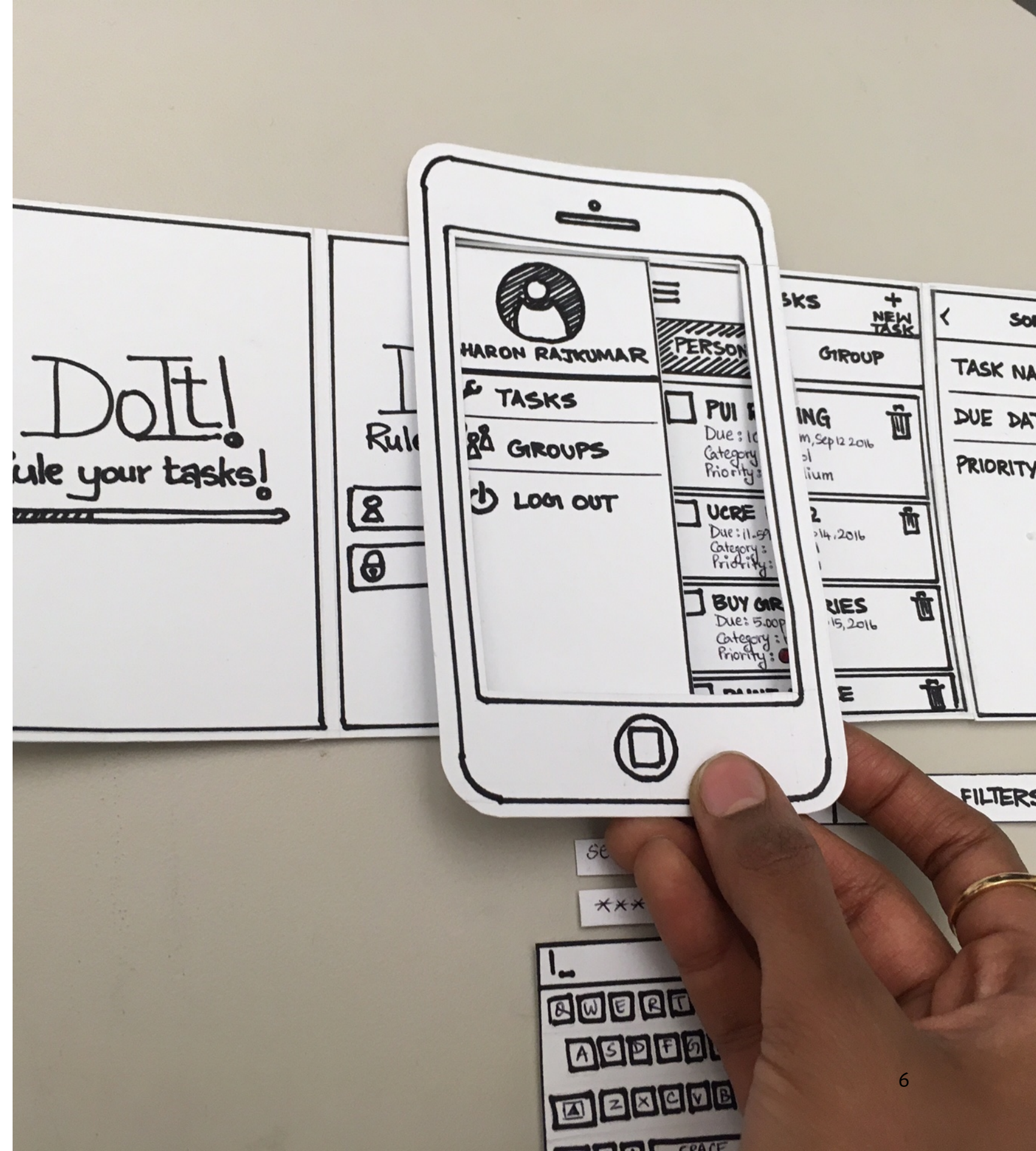
Prototyping Methods

- Paper prototyping
- Wireframes (with Annotation)
- Interactive prototyping
- Native prototyping

Paper Prototyping²

Definition: Mocking up design ideas by sketching pages/screens with design elements using design supplies (e.g., paper, pencils, markers, scissors, glue, tape) and simulating how the envisioned system would respond to user input by swapping different pages/screens and moving/changing design elements.

²Image source



³Image source: Left top, Left bottom, Right



- Tasks:
1. Open the EE app
 2. Register and login
 3. Remove the minutes file from your home screen
 4. Place the add \$10 Topup file to your home screen
 5. Re-arrange the files on your home screen

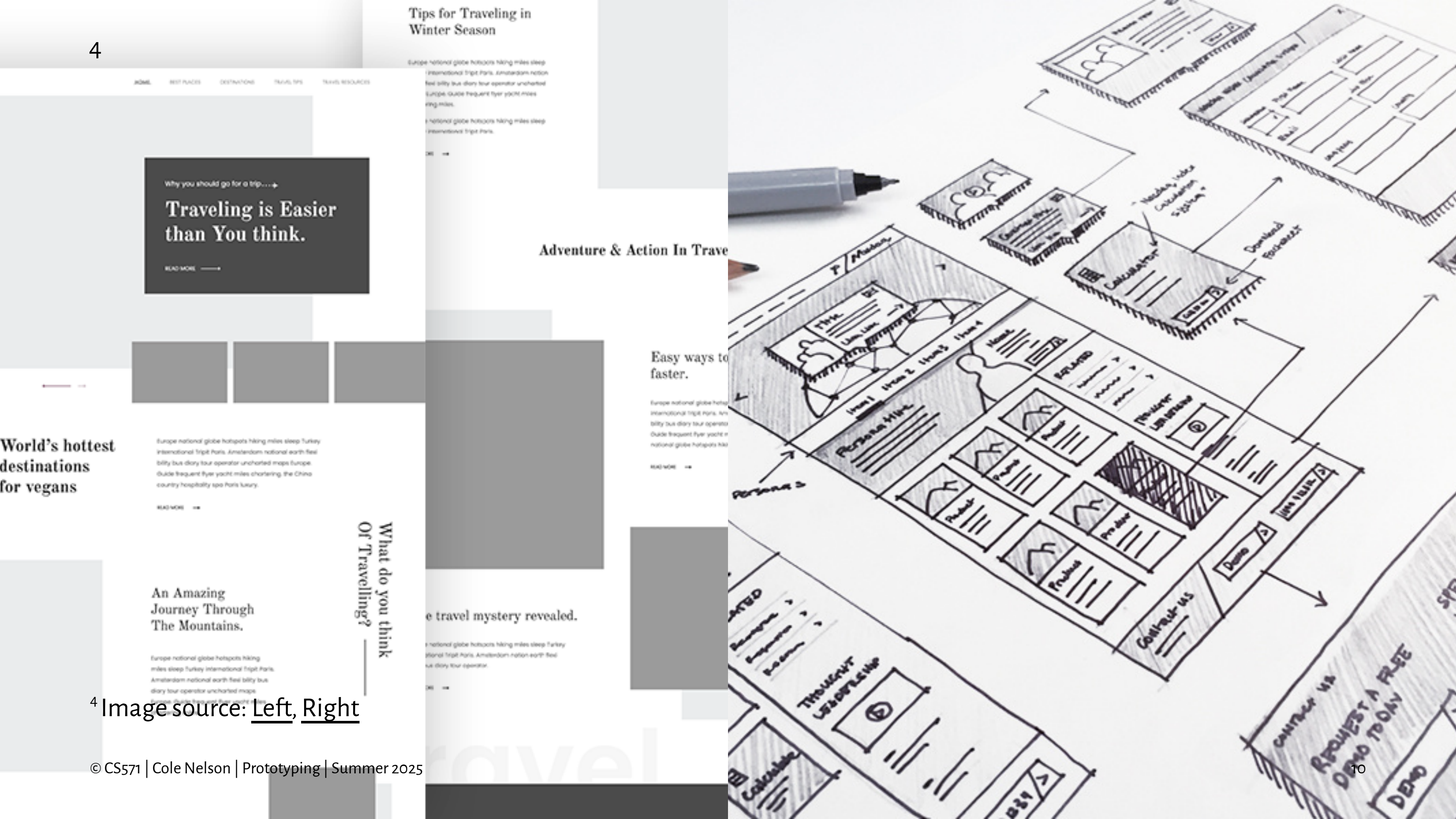
USE NOTIFICATION
INDICATOR
PRESSABLE BUTTON



Wireframes

Definition: Lo-fi prototypes of the makeup of a design in terms of its structural components. Wireframes can be hand-drawn or digitally created.

Most useful in the early-to-mid stages of the design process.



4 Image source: Left, Right

5

Small

Option 1 Option 2 Option 3

Small Medium Large

Small Medium Large

Small Medium Large

5 Image source: Left, Right

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graph TD; JB[John Battelle  
Federated Media Publishing Inc.] --- CB[Carol Bartz  
Yahoo!]; CB --- TO[Tim O'Reilly  
O'Reilly Media, Inc.]; TO --- ES[Eric Schmidt  
Google]; ES --- EW[Evan Williams  
Twitter, Inc.]; JB --- JB2[John Battelle  
Batelle]; ES --- ES2[Eric Schmidt  
Google];
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John Battelle
Federated Media Publishing Inc.

Carol Bartz
Yahoo!

Tim O'Reilly
O'Reilly Media, Inc.

Eric Schmidt
Google

Evan Williams
Twitter, Inc.

John Battelle
Batelle

Eric Schmidt
Google

er 2025

February 26 2014

March 27 2015

April 28 2016

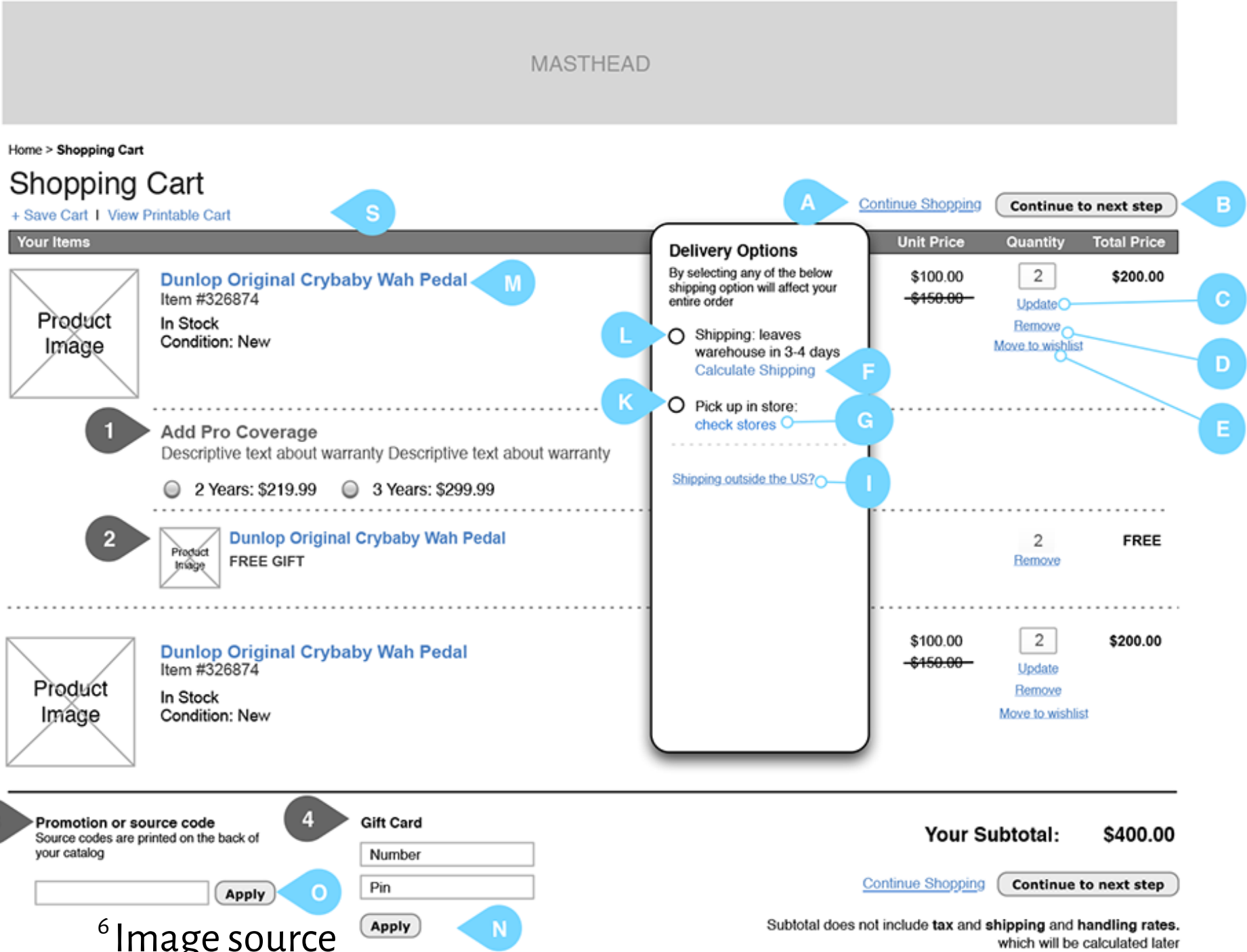
Cancel Done

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Annotations

Definition: Labels, explanations, and notes that provide further information on the goals, content, and functioning of the design elements illustrated on wireframes.

Key in addressing the problem of interpretability of simplified designs for all stakeholders.



Requirements & Description Blocks

Item	Title	Description
1	Add Pro Coverage	When the user selects a specific pro coverage option, it will automatically be added to the cart and the line will collapse – See Page 7
2	Free Gift	The associated product should visually look as part of the parent product.
3	Promo or Source Code	Please refer to Page 7
4	Gift Card	Please refer to Page 7

Field Label	Field Name	Type	Description	Field Length	Read Only (Y/N)
A	Continue Shopping	Text Link	Back to previously viewed page	n/a	N
B	Continue to next step	Button	Go to Checkout Page. After checkout redesign, got to sign In page	n/a	N
C	Update	Text Link	Updates line item quantity.	n/a	N
D	Remove	Text Link	Remove line item from cart	n/a	N
E	Move to wishlist	Text Link	Removes the item from the cart and adds the item to wish list. No changes to the current wish list process	n/a	N
F	Calculate Shipping	Text Link	As is today – No Change	n/a	N
G	Check Stores	Text Link	Direct users to Pick up in store page (Page 3) For public site, use the users location determined by Geo IP to show the stores in the order of proximity. If from SPO, use the store's actual location by default User can input a zip code in the search bar above to change location Please sync up this overlay and the one used on Product detail so that they are identical Page functionality is the same as the one on Product Detail, except we have made the following changes	n/a	N

Email Validation & Confirmation

What Email Would You Like to Send It To?

roger2453@gma

Oops! The email you entered might not be a valid one. Double check it again just to be sure.

Backspace

@#

.

.com

Send

Invalid
"Oops! The email you entered might not be a valid one. Double check it again just to be sure"

Invalid
"Oops! The email you entered might not be a valid one. Double check it again just to be sure"

TAP

Success State

Successfully Sent!

Failure State

Re-Send

Something went wrong

Sending State

Sending...

Get Assistance

Get Assistance

Start Over

Exit

Currently Not In Stock

Get Assistance

Help is on the way. If we get too busy, go ahead and grab one of us!

Remains for 10 seconds

TAP

TAP

Warning Modals

Which Colors Would You Like to See?

Wait!

If you change the model, we won't be able to keep your existing colors and packages.

Is that okay?

No

Yes

Continue

Other Modals

Wizard AND Filter Change Model Event

- When user tries to change the model toggle on results screen (selecting an actual model button, not the expand down button)

"If you change the model, we won't be able to keep your existing colors, packages, or model year. Is that okay?"

Start Over Event

"If you start over, you'll lose any work you've done so far. Is that okay?"

Exit Event

"You're about to exit the inventory search and will lose any work you've done so far. Is that okay?"

Internet Down

Which Colors Would You Like to See?

For some reason, our internet is down.

We've already notified someone to fix it. In the mean time, grab one of us and we'll be happy to help you!

Continue

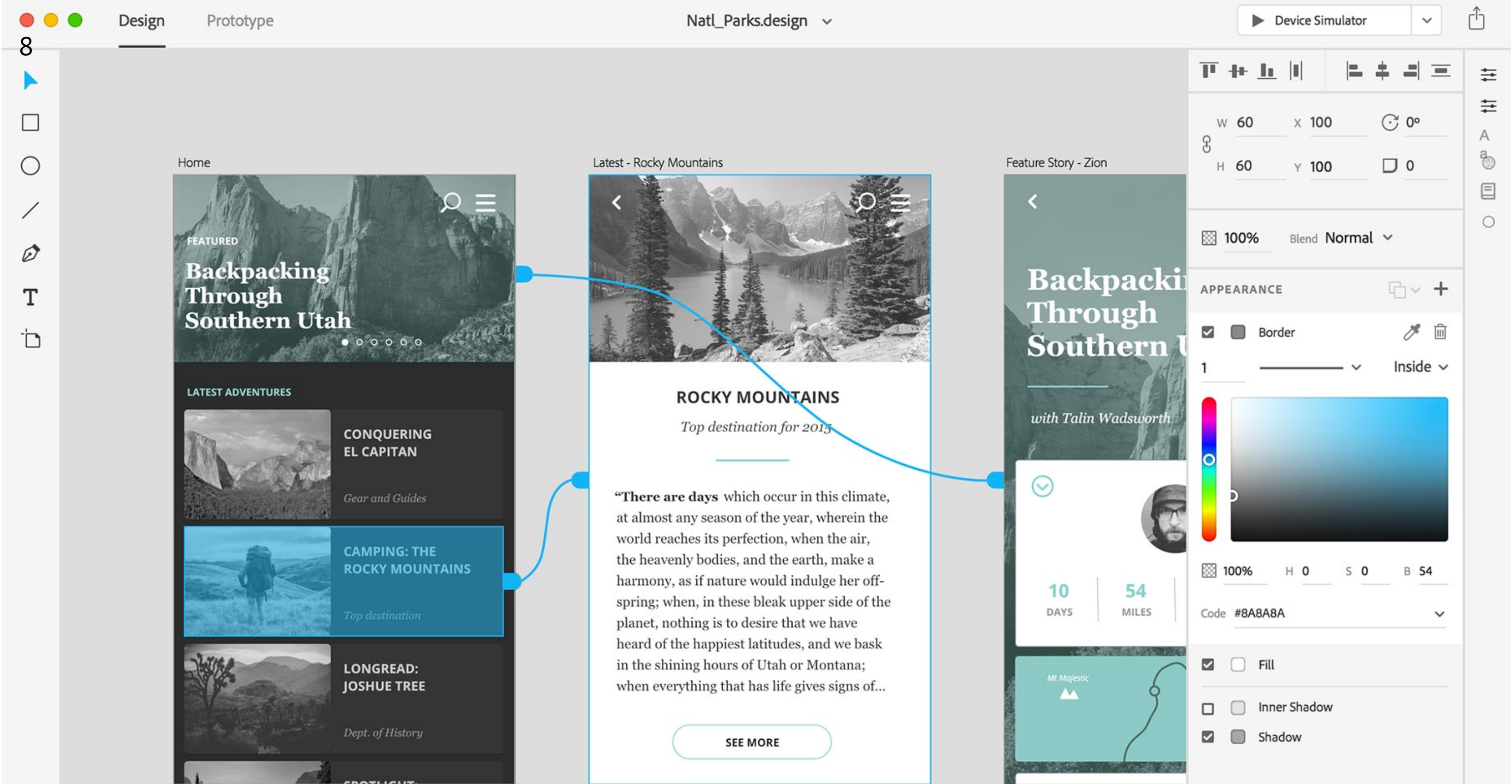
Send an email (primary email address the dealership provides) stating the kiosk is not receiving internet

Filter Changes

Re-adjusting Filters...

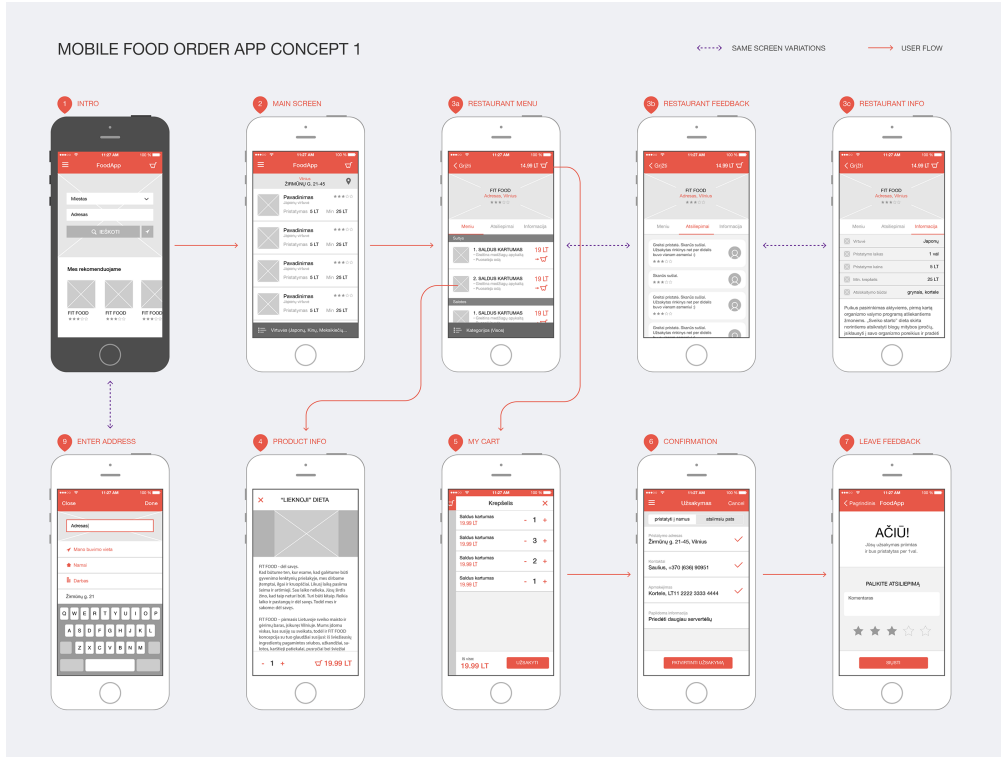
Occurs when model or year is changed

Definition: Creating realistic prototypes of the navigation or structural components (or both) of the design idea by creating a series of screens/pages with design elements, linking these screens/pages for navigation, and simulating the transitions between screens/pages.



⁸ Image source

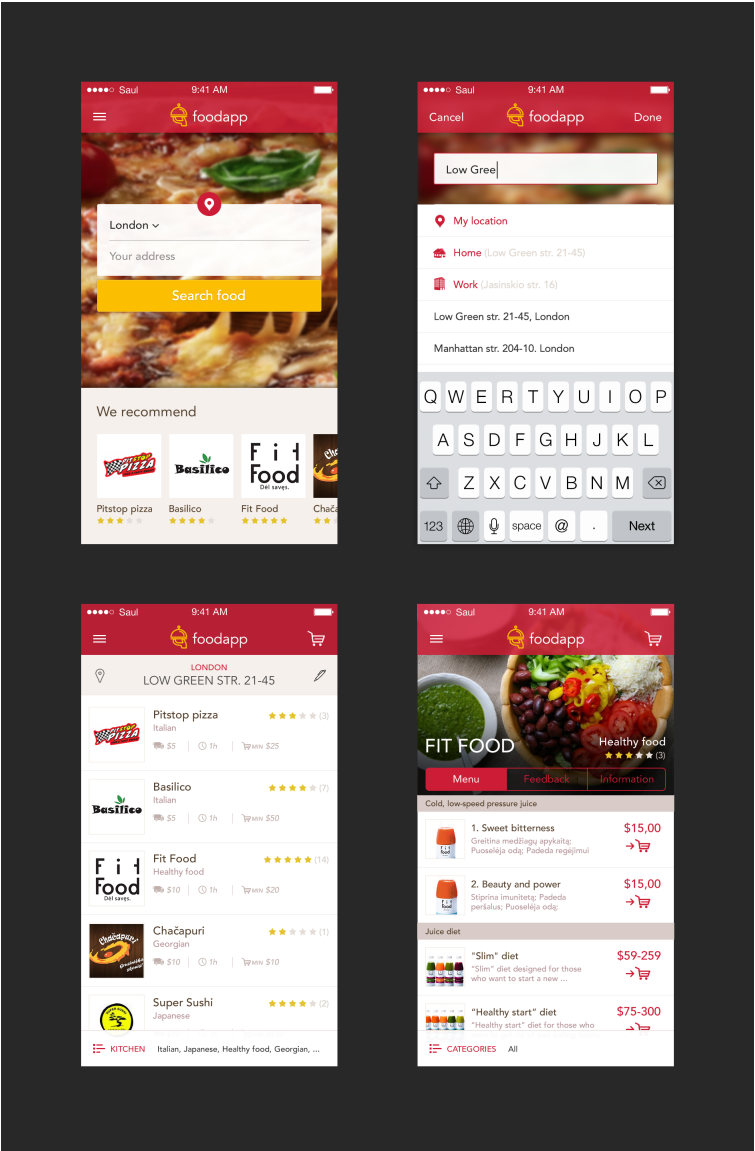
Interactive prototypes can use *lo-fi* or *hi-fi* components.⁹



WIREFRAME



PIXEL-PERFECT MOCKUP



⁹ Images source

ICA: Prototyping

Create a *lo-fi* prototype of Badger Mart for a mobile device!

Re-visit the HW7 Demo video for features...

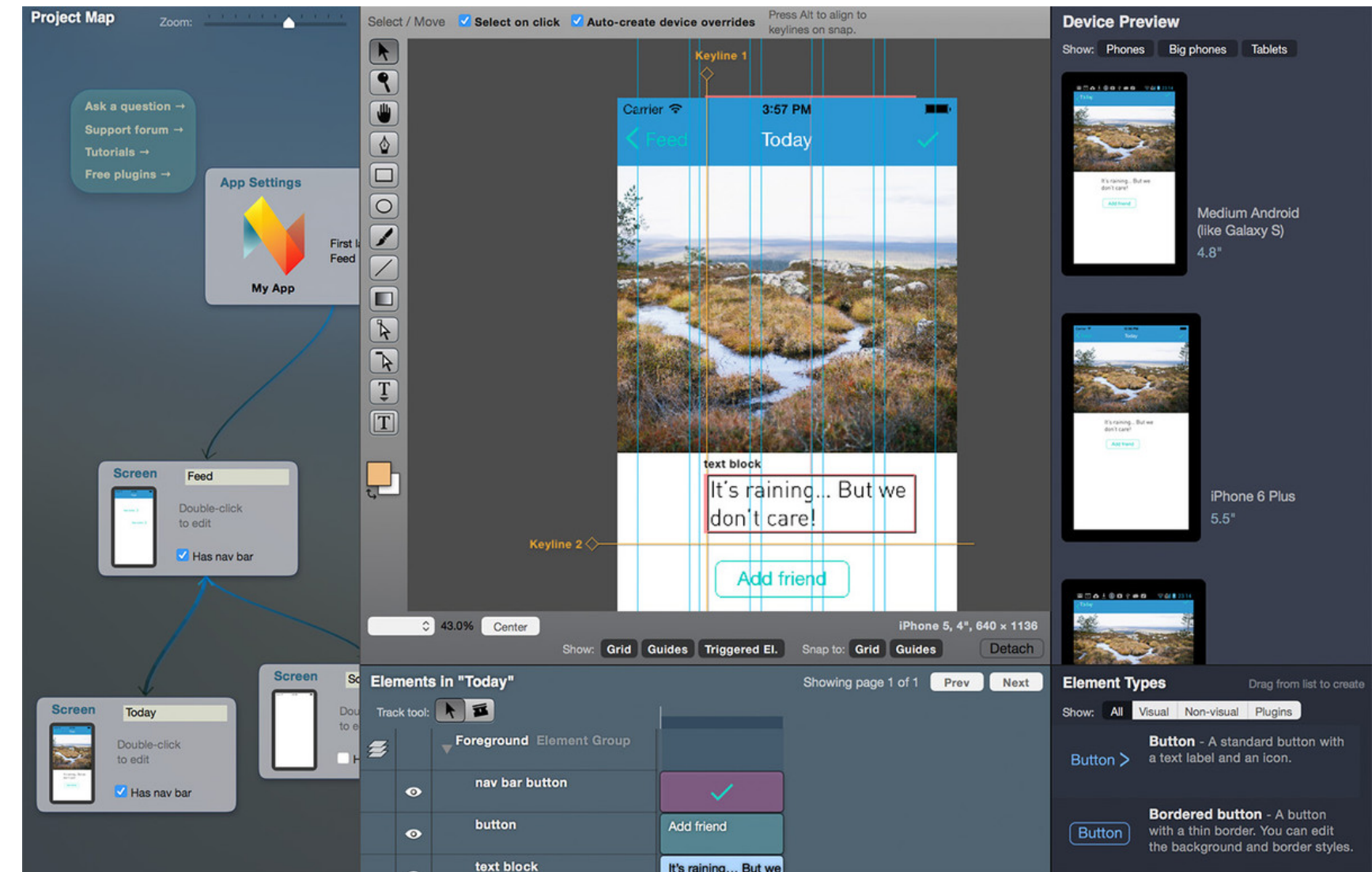
- A user should be able to view available products.
- A user should be able to add and remove products from their basket.
- A user should be able to submit an order.

Native prototyping¹⁰

Definition: Implementing and testing design ideas on the target technology platform of the design.

Examples:

- Neonto (bust!)
- React Studio (bust!)
- Bootstrap Studio

¹⁰ Image source

Prototyping Theory

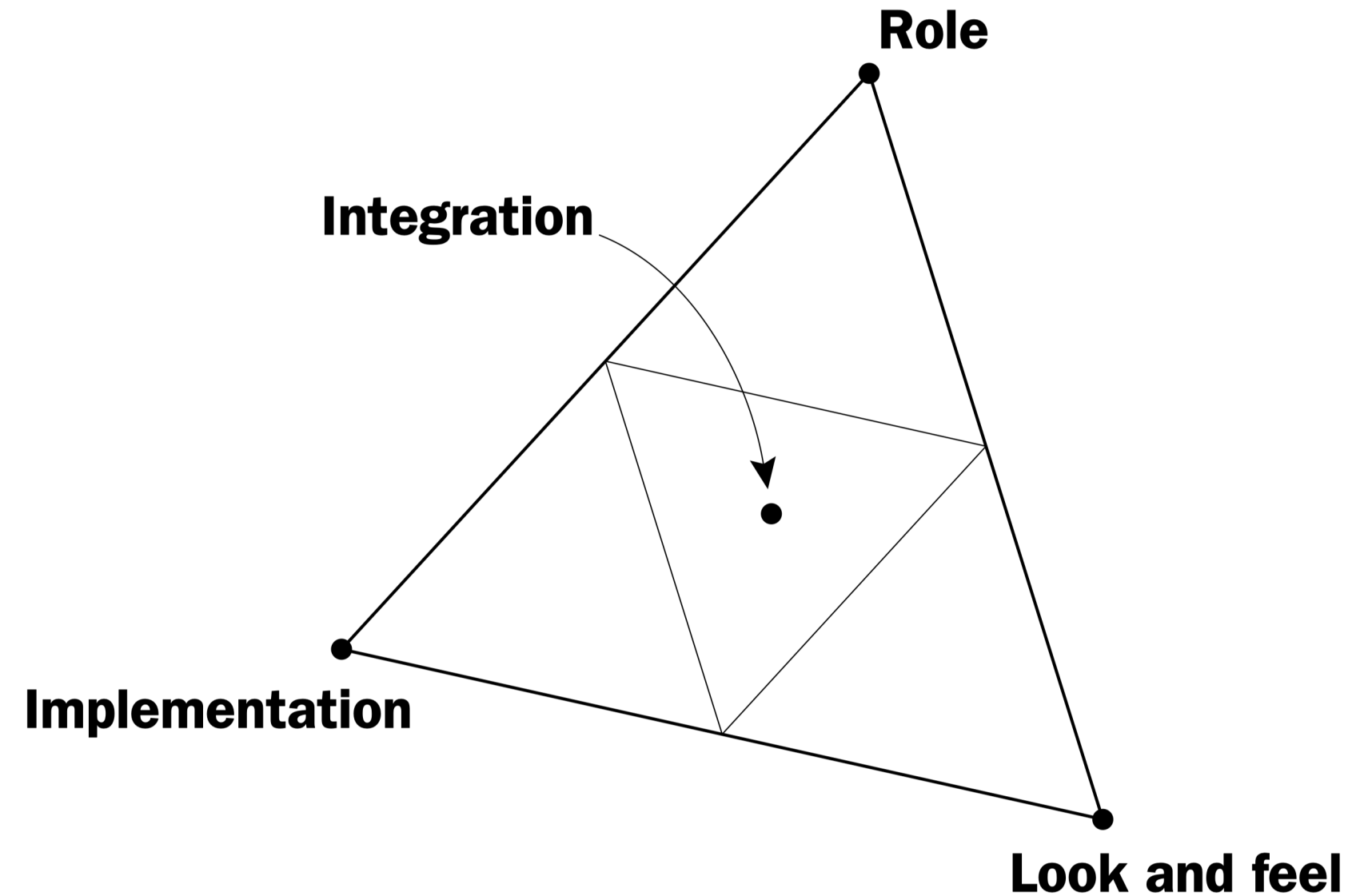
Prototyping Models & Strategies

- Three-dimensional model of prototyping
- Prototyping scope
- Prototyping strategies
- Prototyping fidelity

Three-dimensional Model of Prototyping¹³

Prototypes represent three dimensions of a design idea:

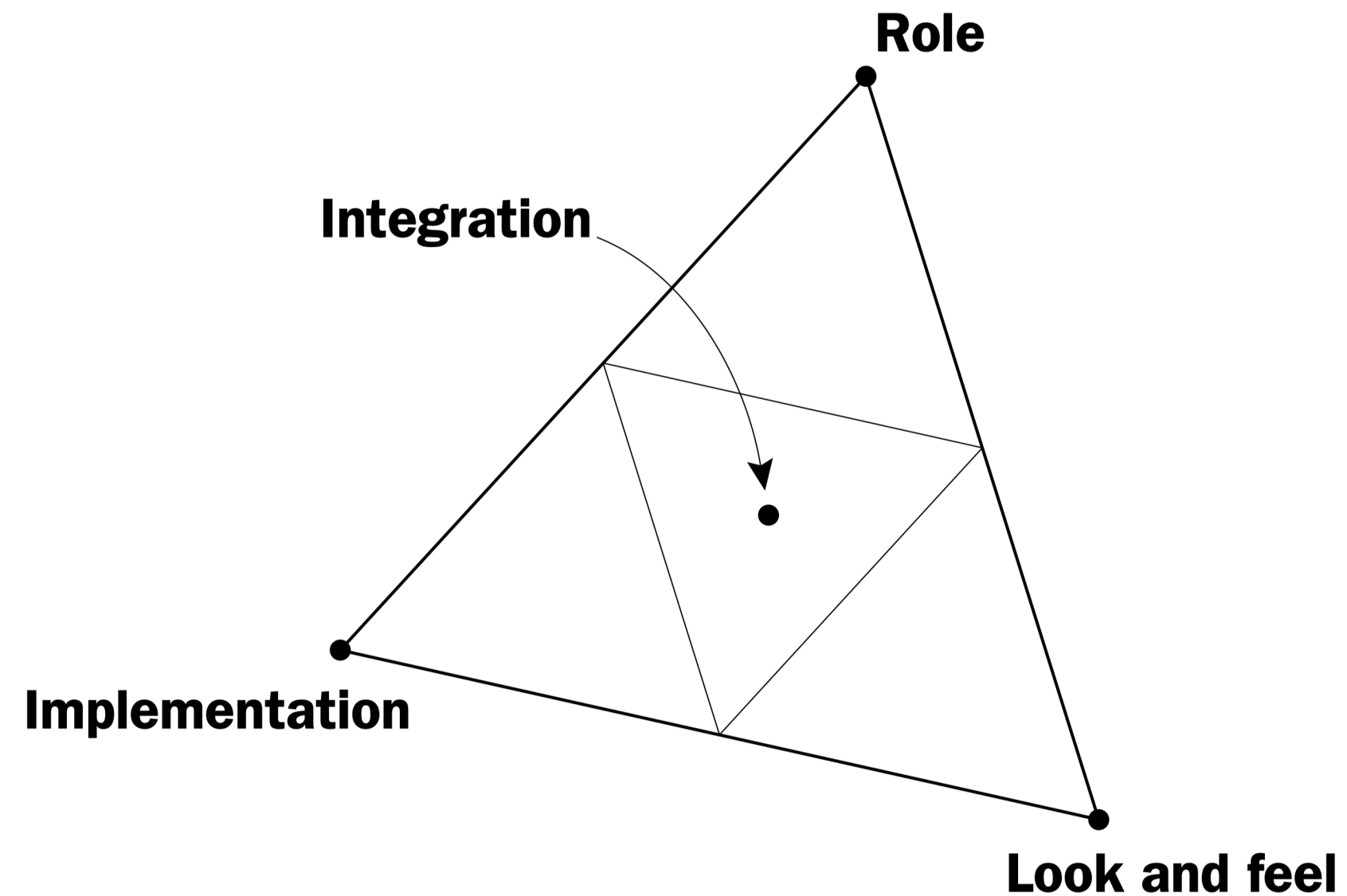
1. Role
2. Look and feel
3. Implementation



¹³ Houde & Hill, 1999. What do prototypes prototype?

Each dimension can be represented at various levels of fidelity.¹³

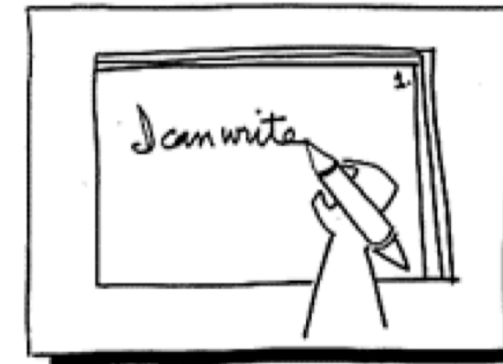
Their integration makes a working prototype or a pre-alpha product.



¹³ Houde & Hill, 1999. What do prototypes prototype?

Dimension 1: Role¹³

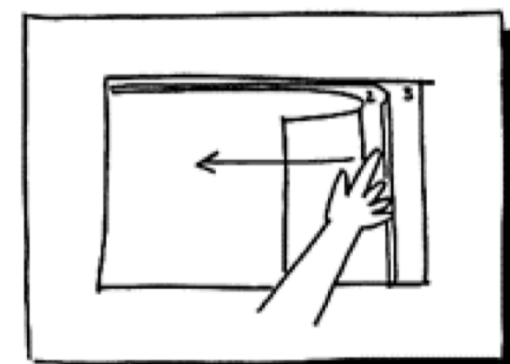
Definition: Represents the functions that the system serves in the user's life, i.e., how the system is useful to them.



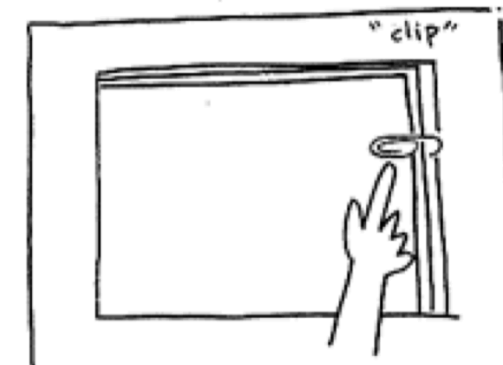
L. can write into her notebook.



She can flip over her pen and erase.



L. can flip to the next page.



L. marks her place.



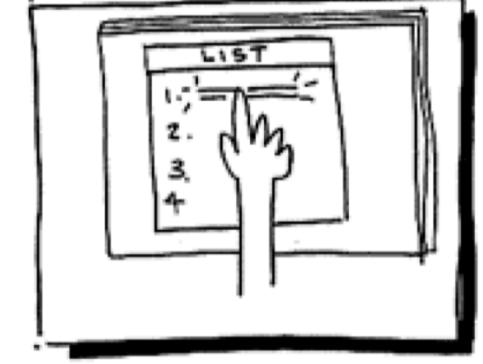
L. marks a phone number.



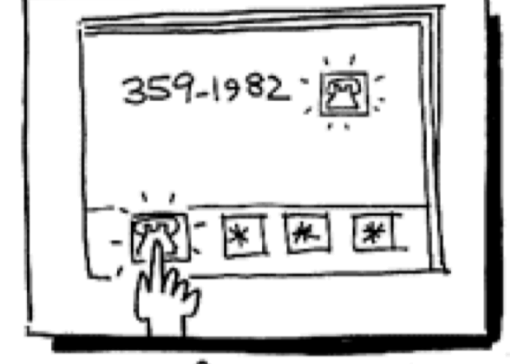
L. shifts to the InBox Section.



L. adds comments



L. brings up a list.

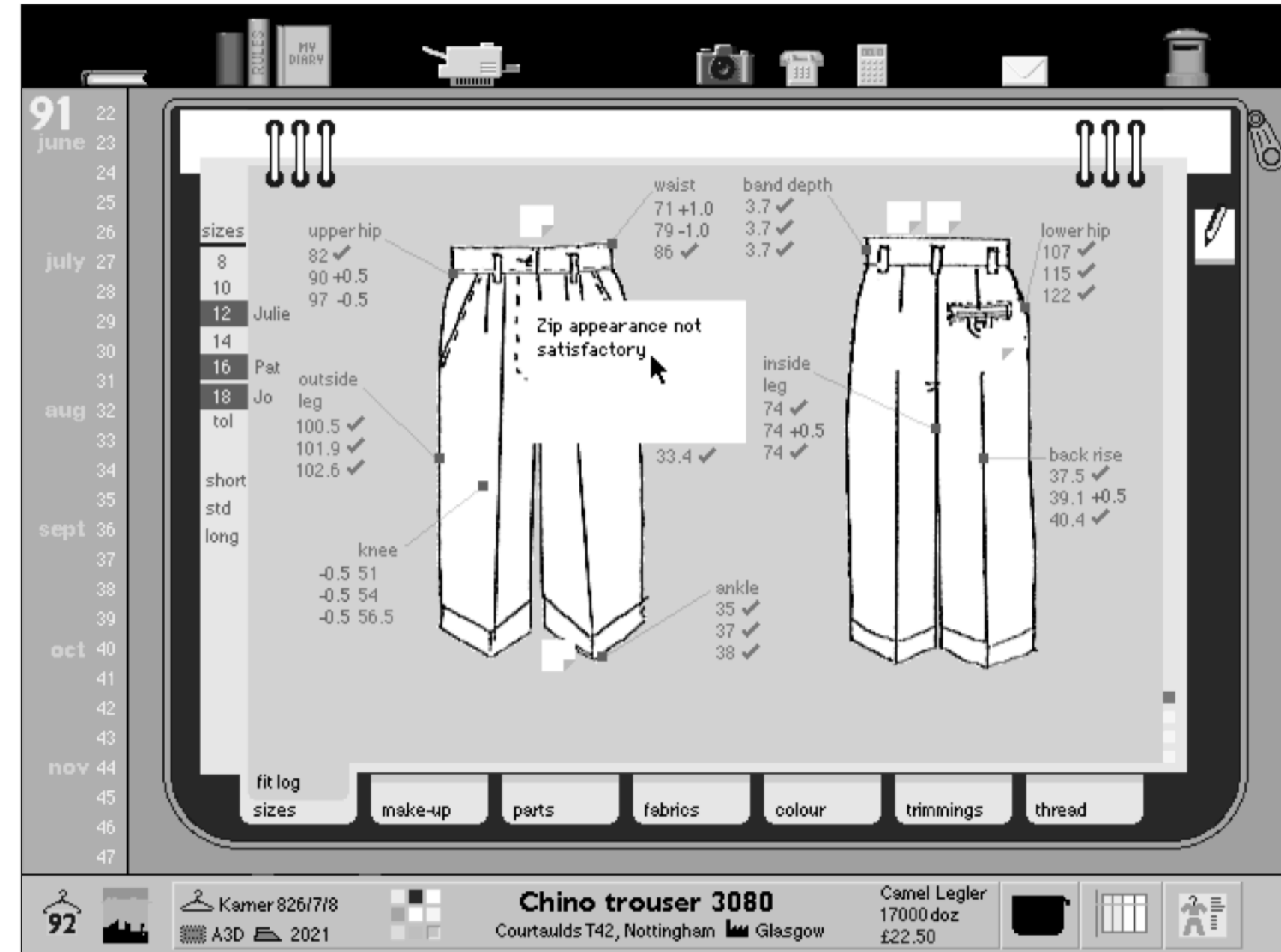


L. looks for phone numbers.

¹³ Houde & Hill, 1999. What do prototypes prototype?

Dimension 2: *Look and Feel*¹³

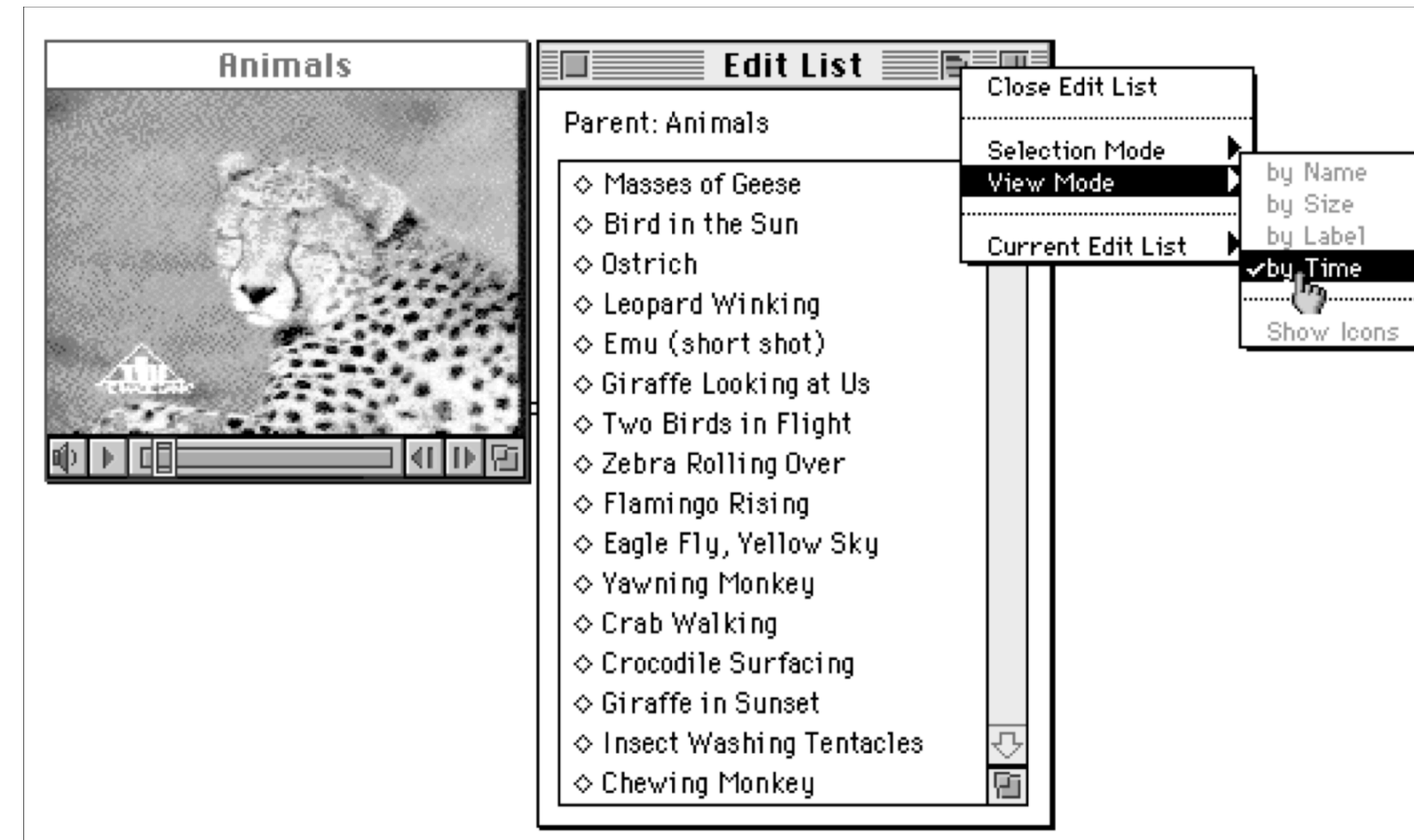
Definition: Simulates the sensory experience of the user while using the system, i.e., what the user sees, hears, and feels during use.



¹³ Houde & Hill, 1999. What do prototypes prototype?

Dimension 3: *Implementation*¹³

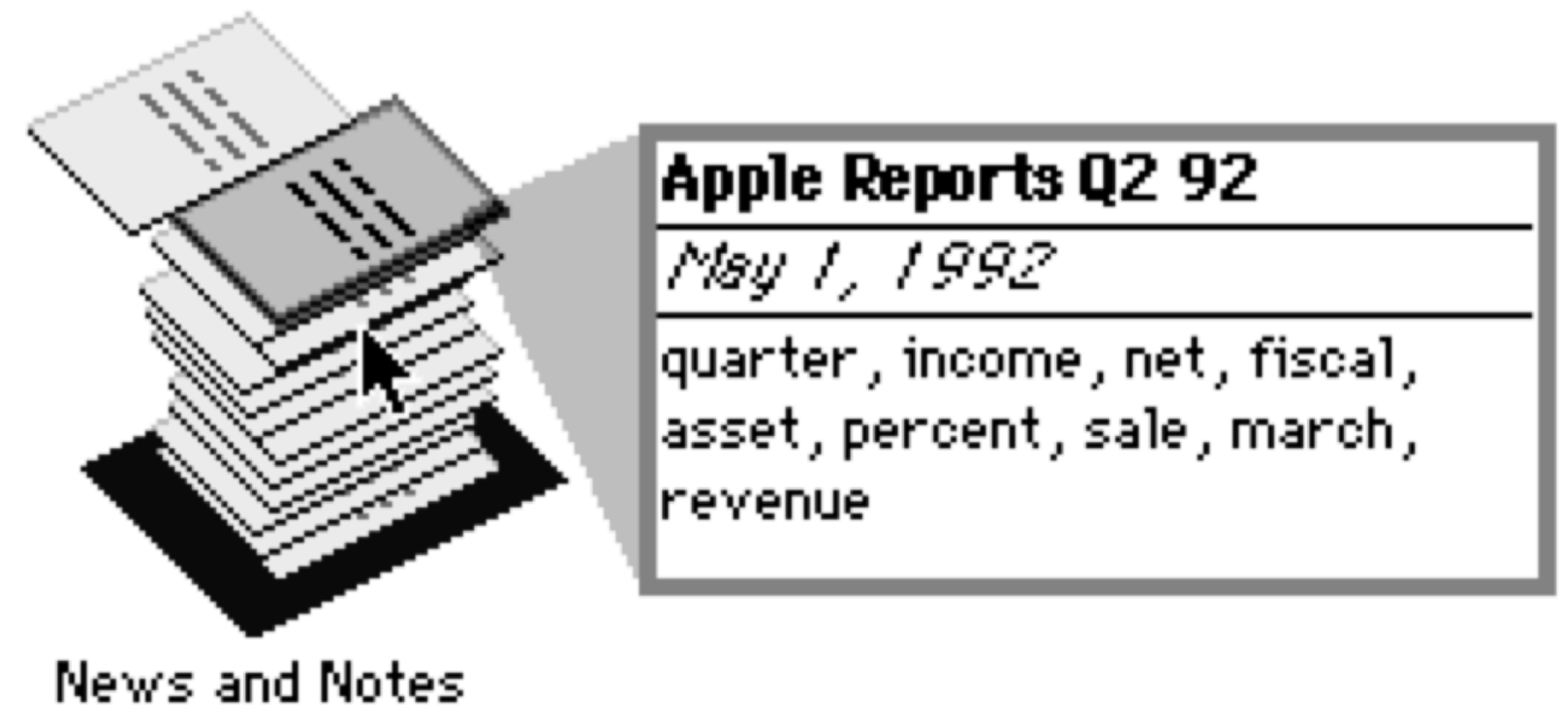
Definition: Includes the technical capabilities that enable the system to perform its function, i.e., the low-level details of how the system works.



¹³ Houde & Hill, 1999. What do prototypes prototype?

Dimensions Combined: *Integration*¹³

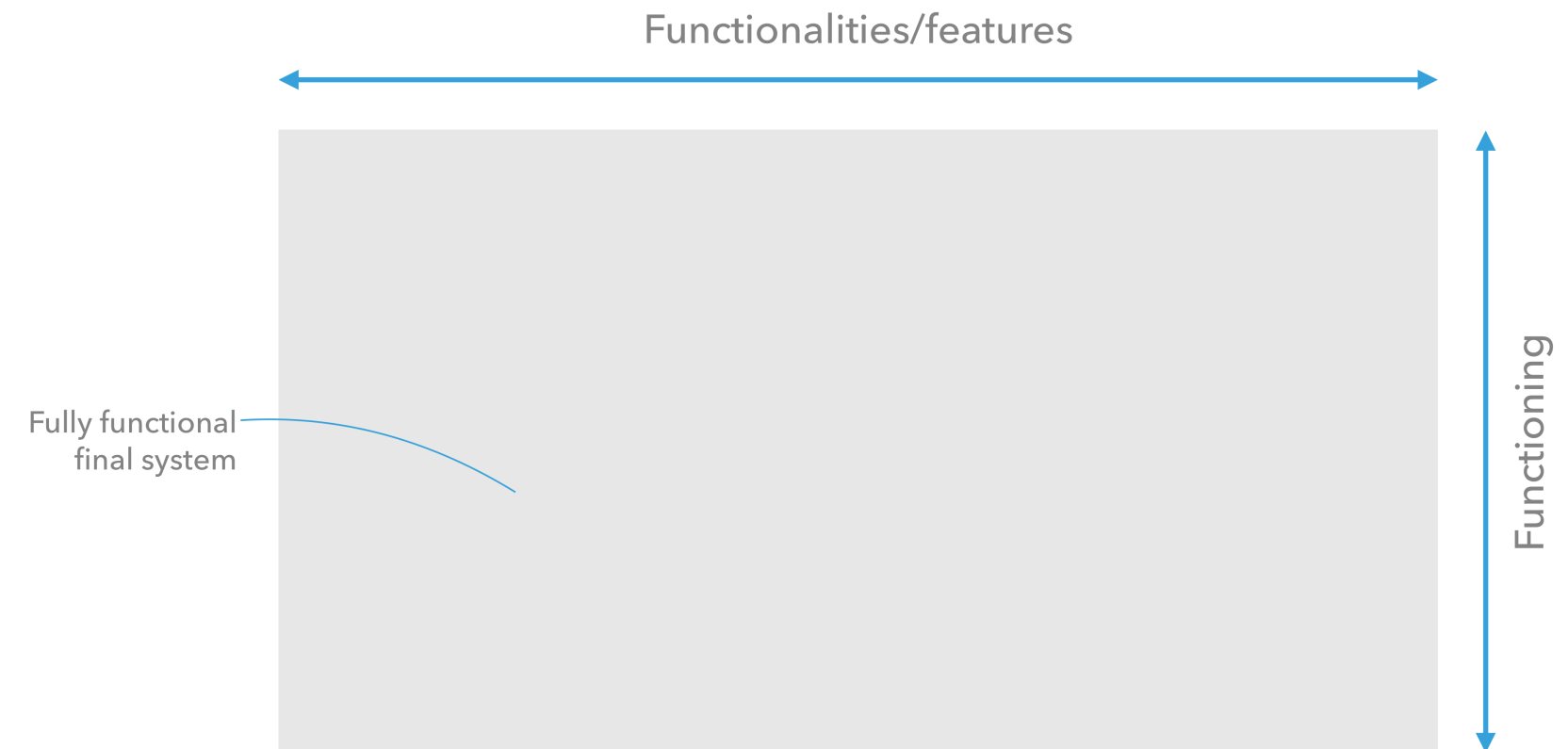
Definition: Represents the complete user experience with the system as envisioned in the conceptual design.



¹³ Houde & Hill, 1999. What do prototypes prototype?

Prototyping Scope

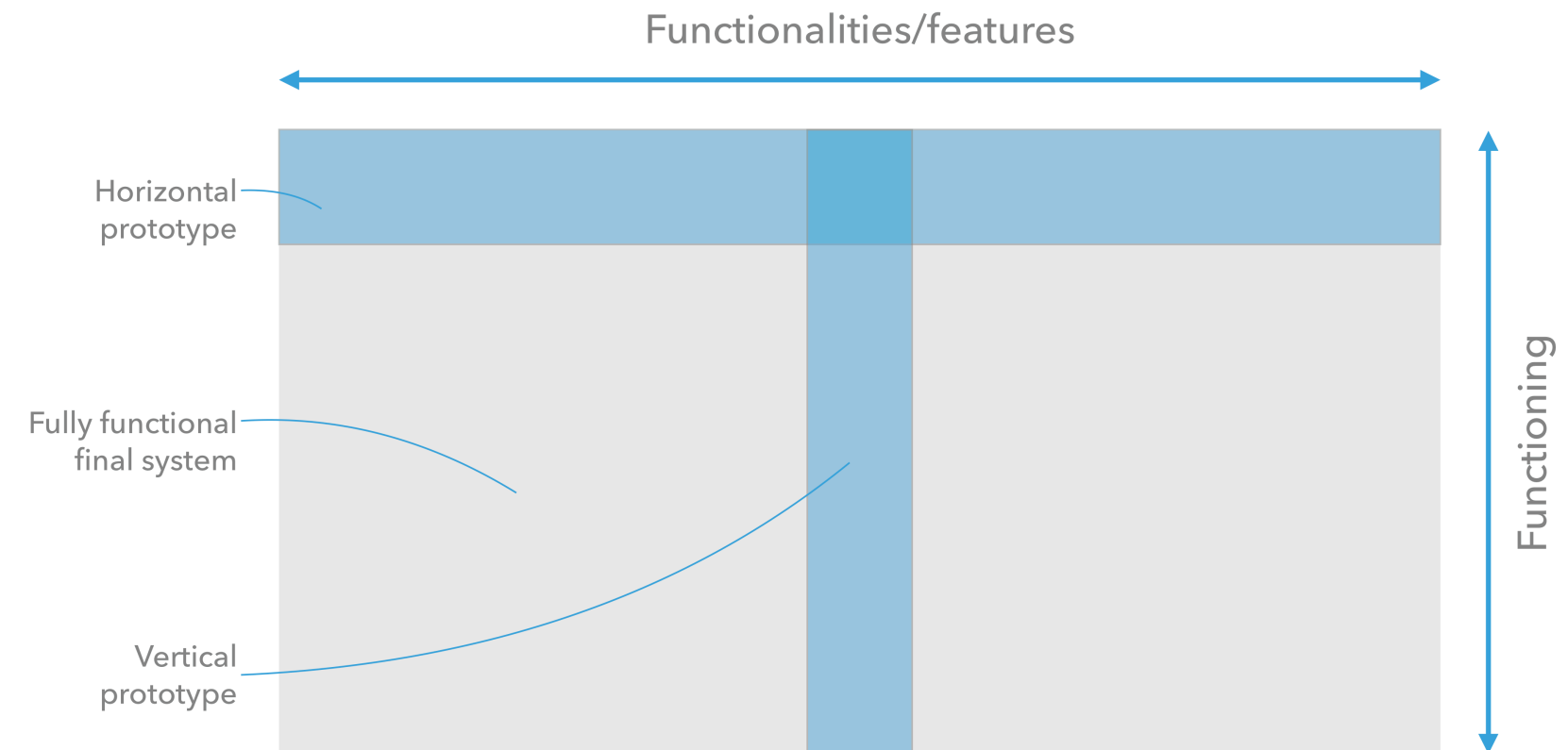
Consider the space of *features* and *functioning* as everything that a system does.



Prototyping Scope, *Continued*

Horizontal Prototype: Provides a broad view of the entire system and focus on the user interaction rather than the functionality.

Vertical Prototype: Focuses on a single feature/functionality and provides the full functioning of that feature.



Prototyping Strategies

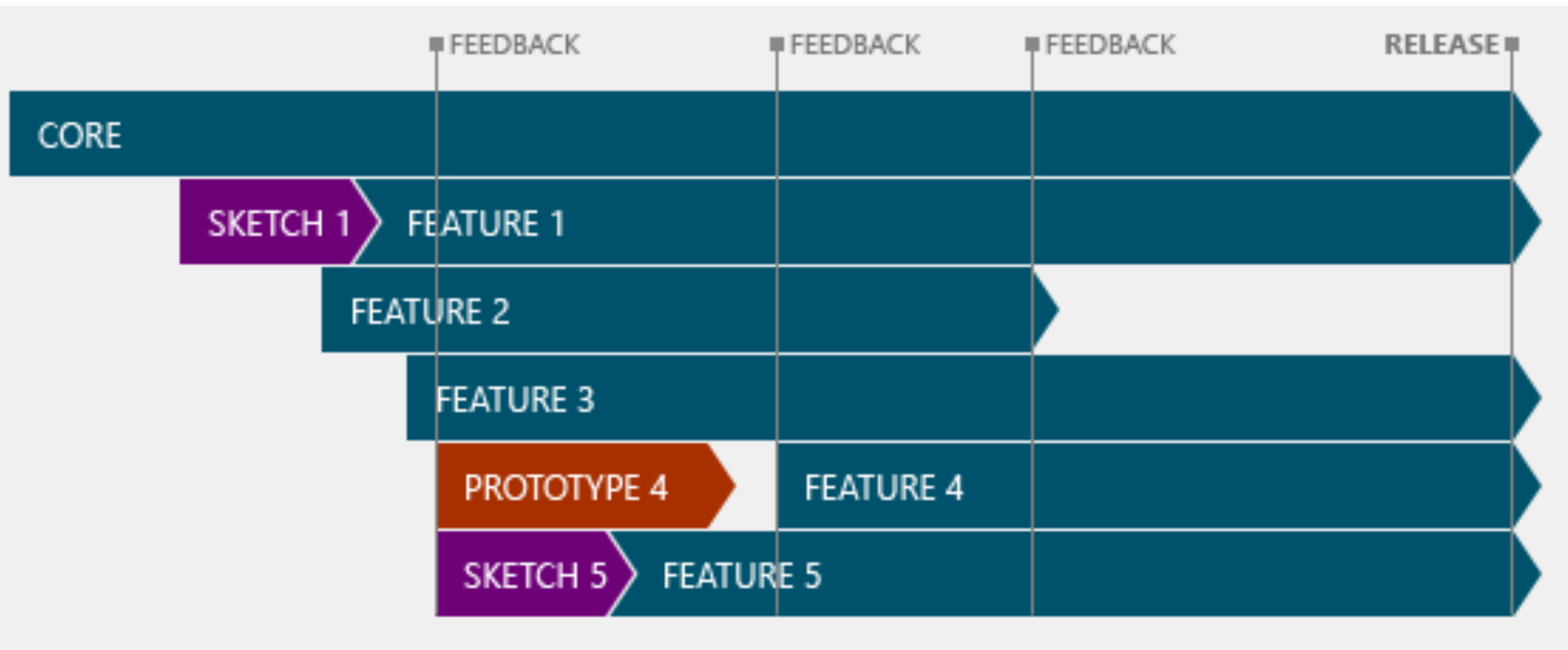
- Throwaway prototyping
- Evolutionary prototyping
- Incremental prototyping
- Extreme prototyping

Throwaway prototyping

Definition: Rapid prototyping to explore design ideas, demonstrate feasibility, communicate with stakeholders, and test the ideas with users and eventually discarding the prototype instead of further developing the model into a final product.

Most lo-fi and paper prototypes are throwaway prototypes.

Throwaway prototyping is usually combined with *sketching*.

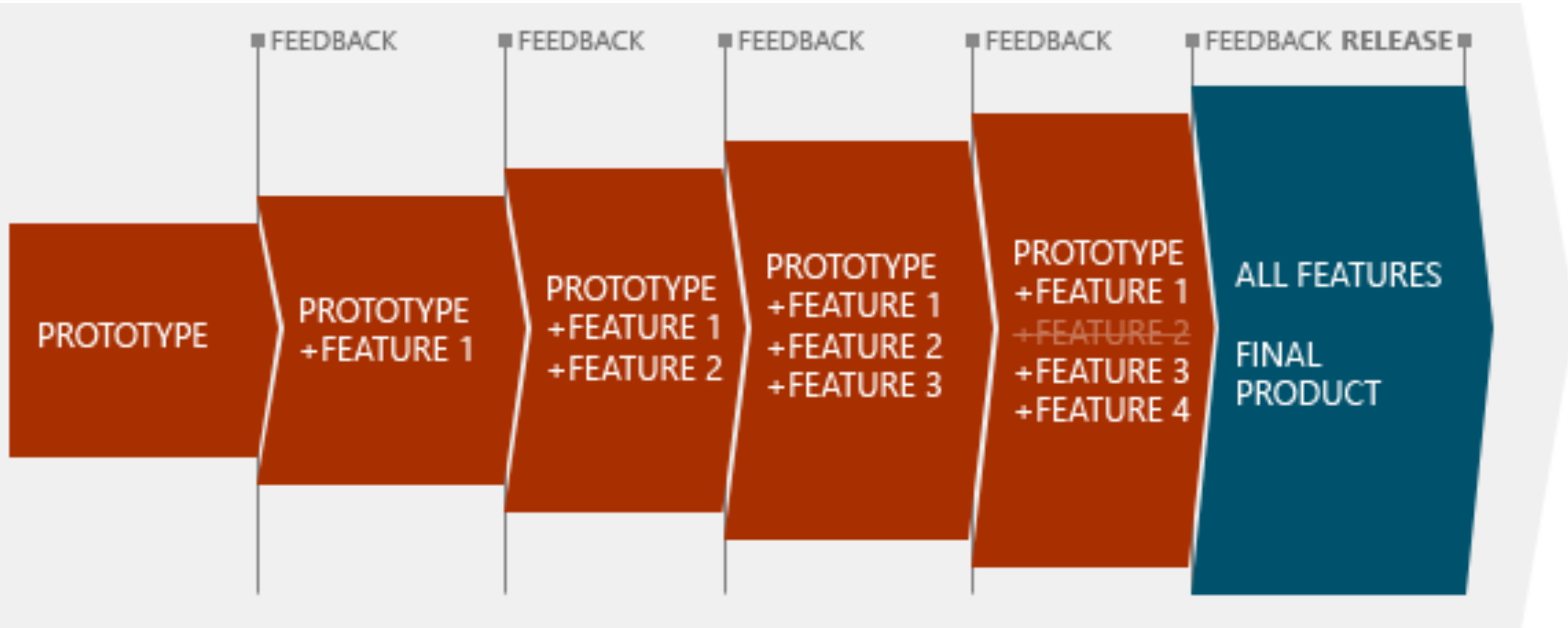


¹⁴ Image source

Evolutionary prototyping

Definition: Also called *breadboard prototyping*, the design team incrementally builds prototypes of a design idea, tests the idea with users, and refines the prototype until it reaches the desired level of maturity.

Most products we use are evolutionary prototypes that went through alpha, beta, etc. phases.

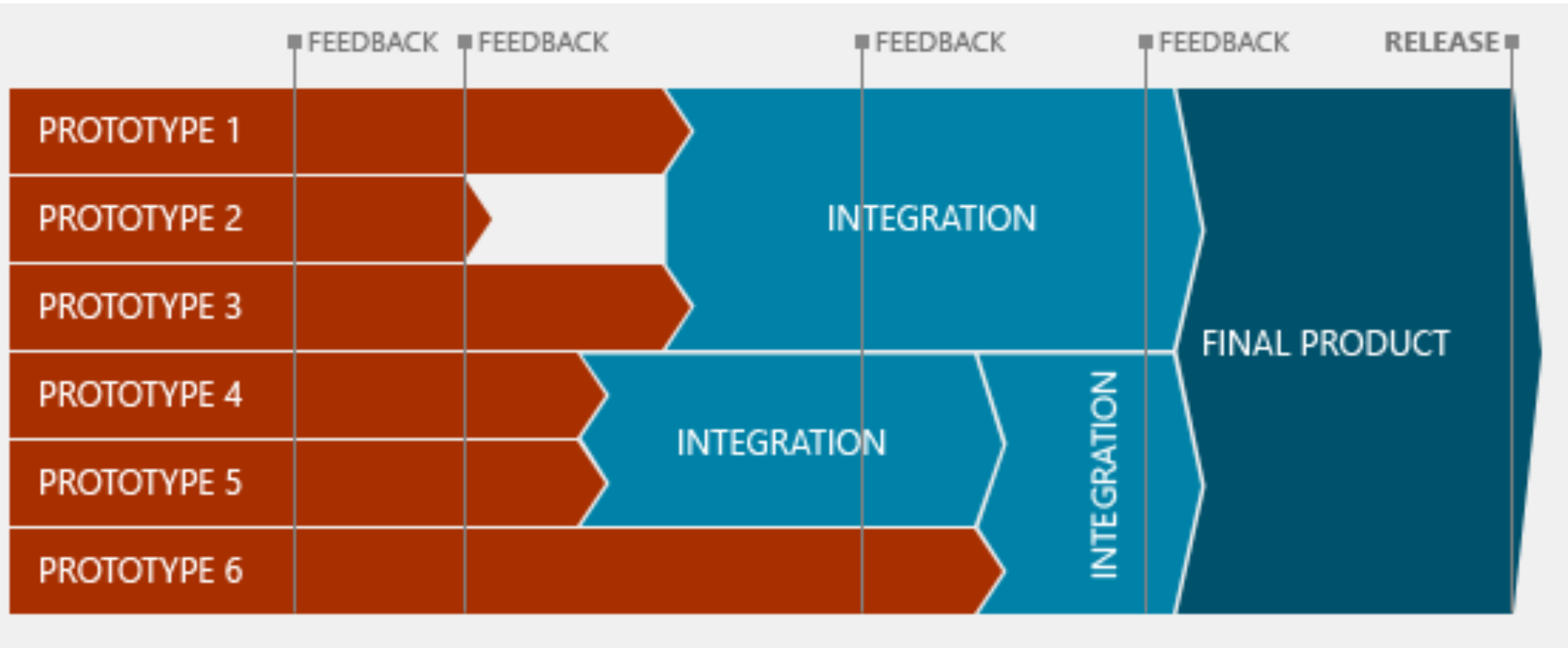


¹⁴ Image source

Incremental prototyping

Definition: Dividing system functionality into slices (vertical prototypes) based on design specifications and incrementally building each slice that is then integrated into the overall system.

Appropriate for large and complex projects.

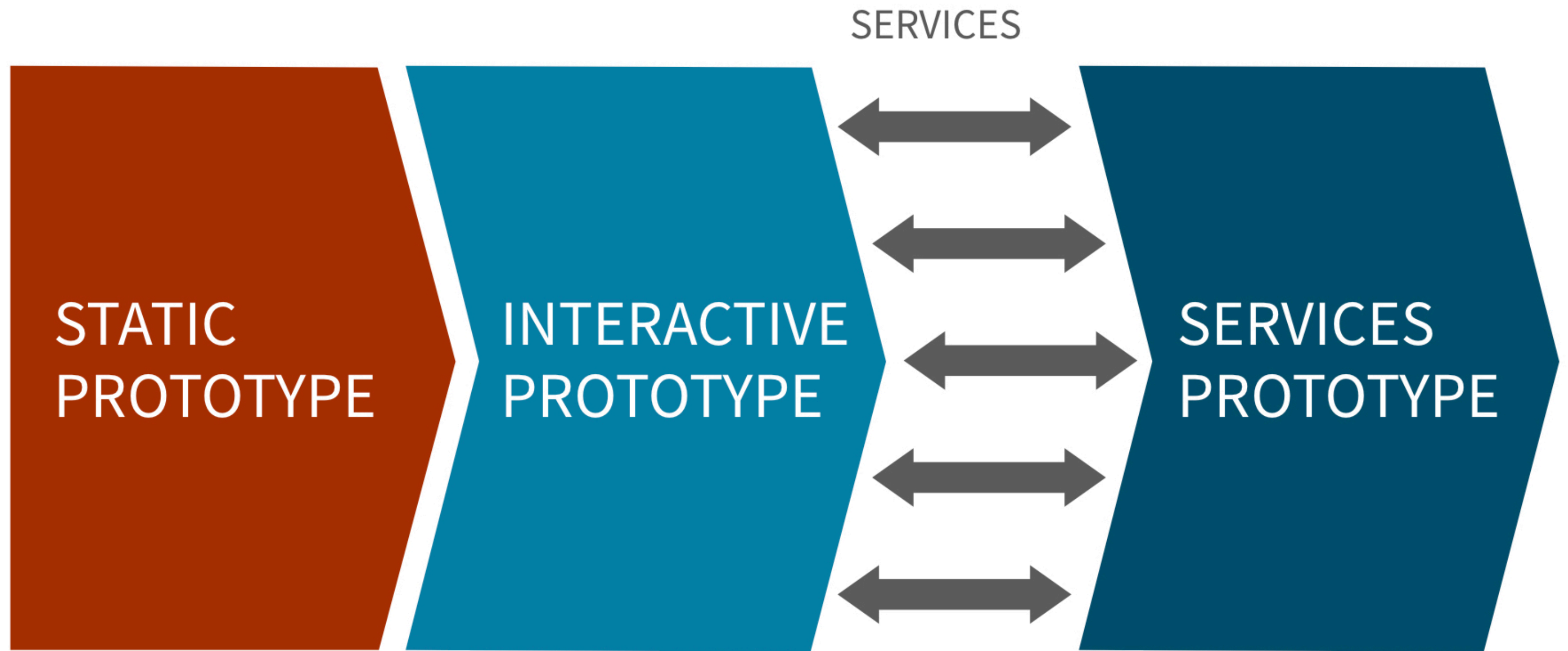


¹⁴ Image source

Extreme prototyping

Definition: Breaking down the development into three phases that build on each other: (1) building a static prototype, (2) building fully functional, interactive components that will simulate services, and (3) finally implementing the services.

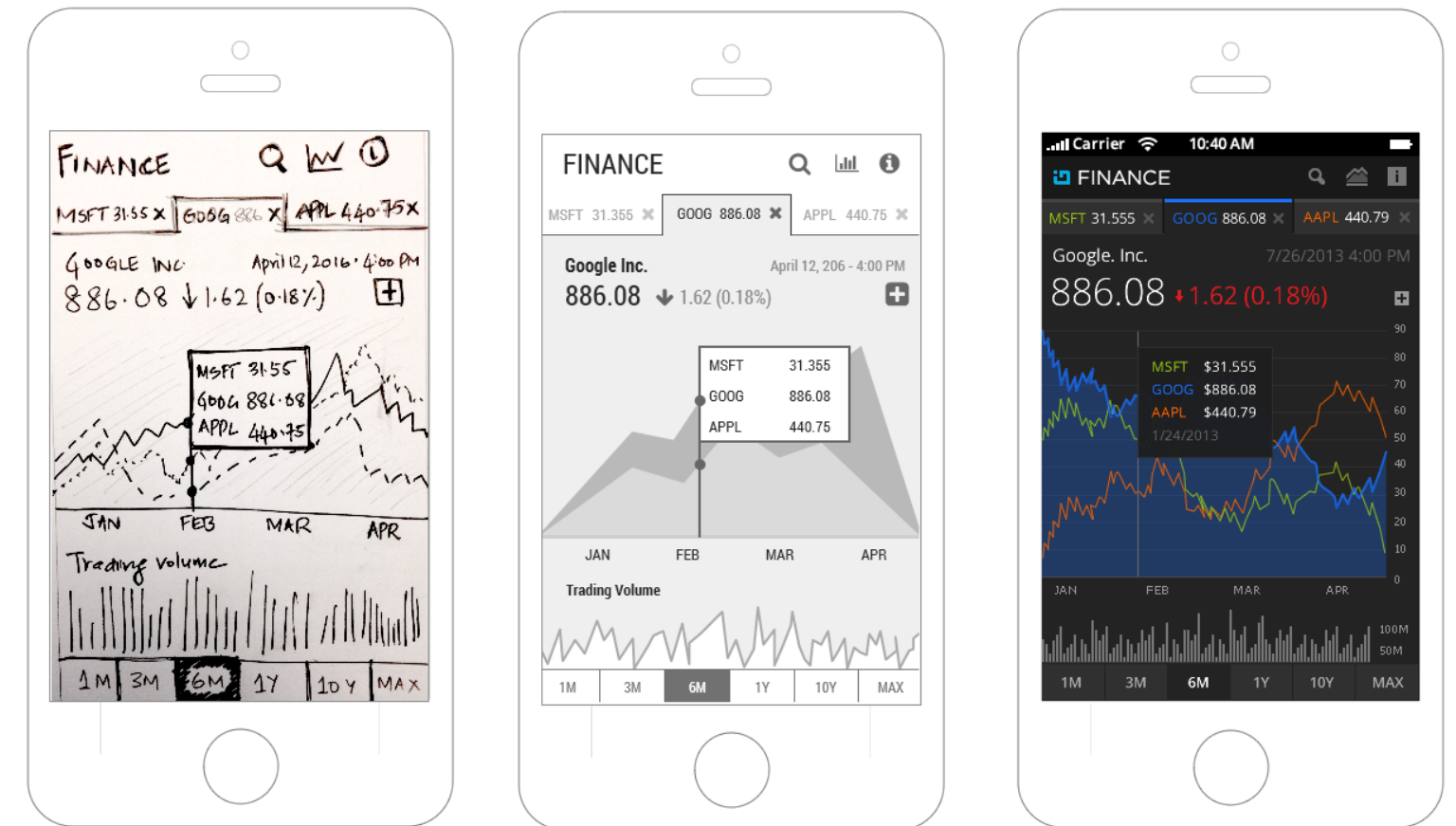
Enables rapid and parallel prototyping, testing, and refinement by removing dependencies between different components of a system or between the system and third party services.



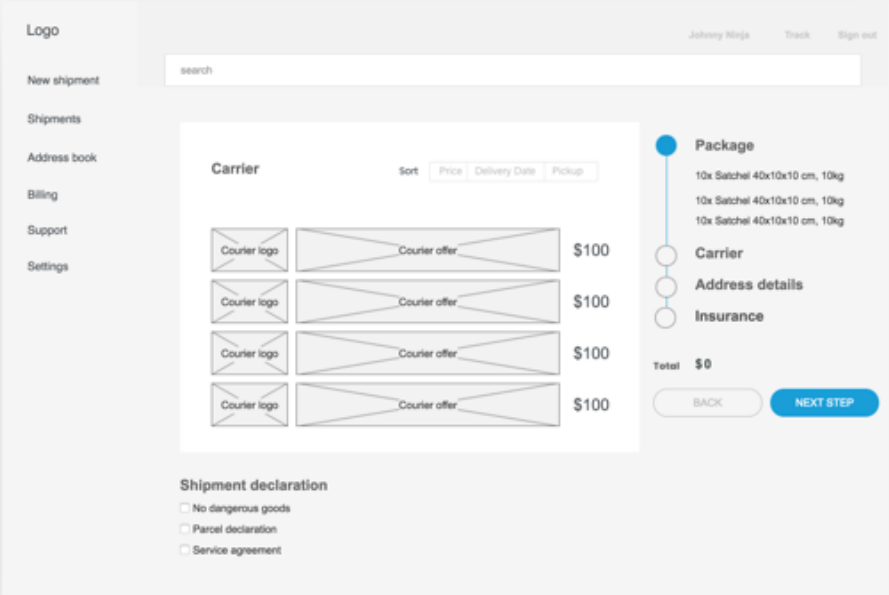
Fidelity in Prototyping¹⁵

Definition: The level of detail in which a design is represented in the prototype.

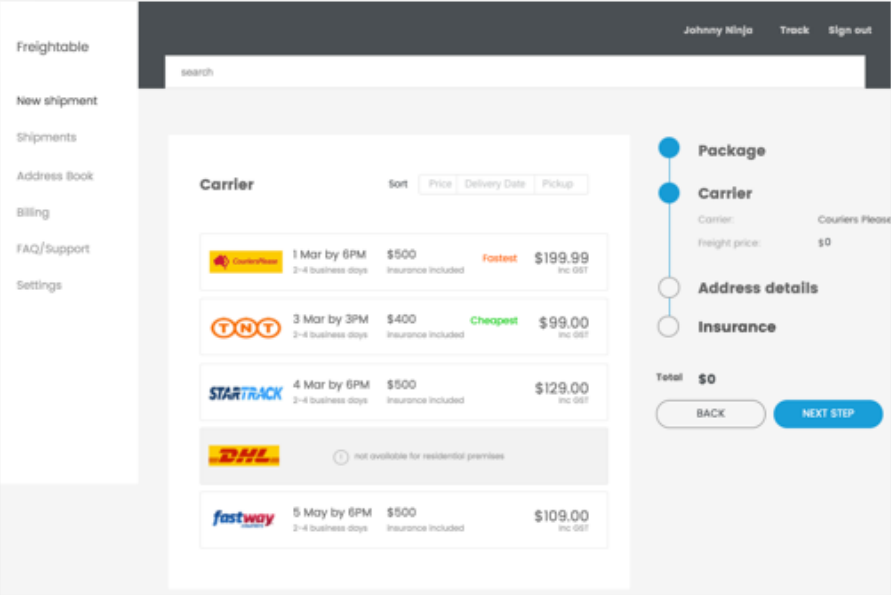
- Two ends of a fidelity spectrum:
 - Low-fidelity (lo-fi) prototyping
 - High-fidelity (hi-fi) prototyping



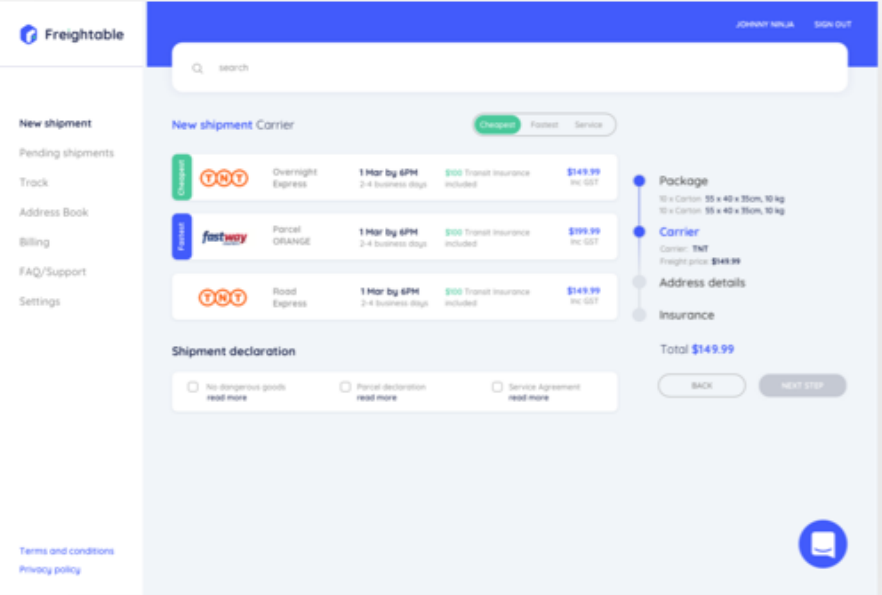
¹⁵ Image source



Low-fidelity



High-fidelity



Final design

¹⁶ Image source

Why is fidelity important?

The more "done" the prototype looks, the narrower the feedback will be; and vice versa.

This is just an idea. I'm not done!

As a principle, use no higher resolution than is necessary.



Why bother with lo-fi prototyping?

- Has lower development cost
- Prevents designers from prematurely wedding to specific design ideas
- Enables exploring, communicating, and testing of conceptual designs
- Helps designers identify structural, navigation, and organizational issues
- Allows rapid evaluation of multiple design ideas

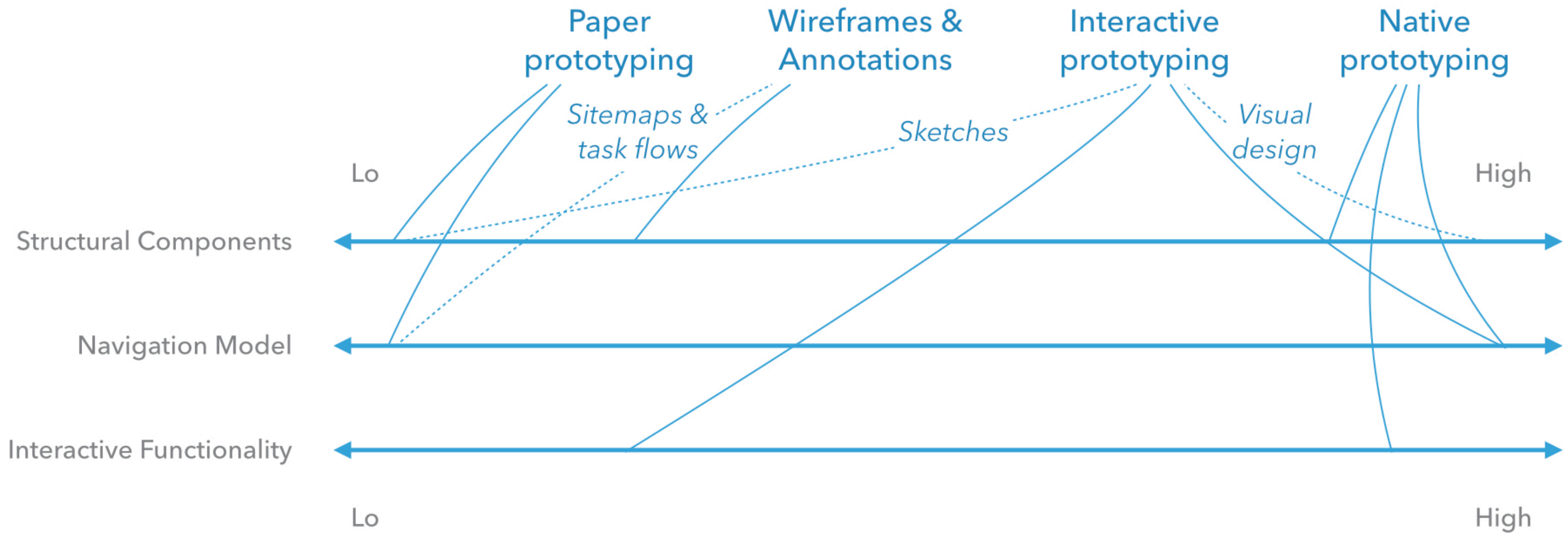
Limitations of Lo-fi Prototyping

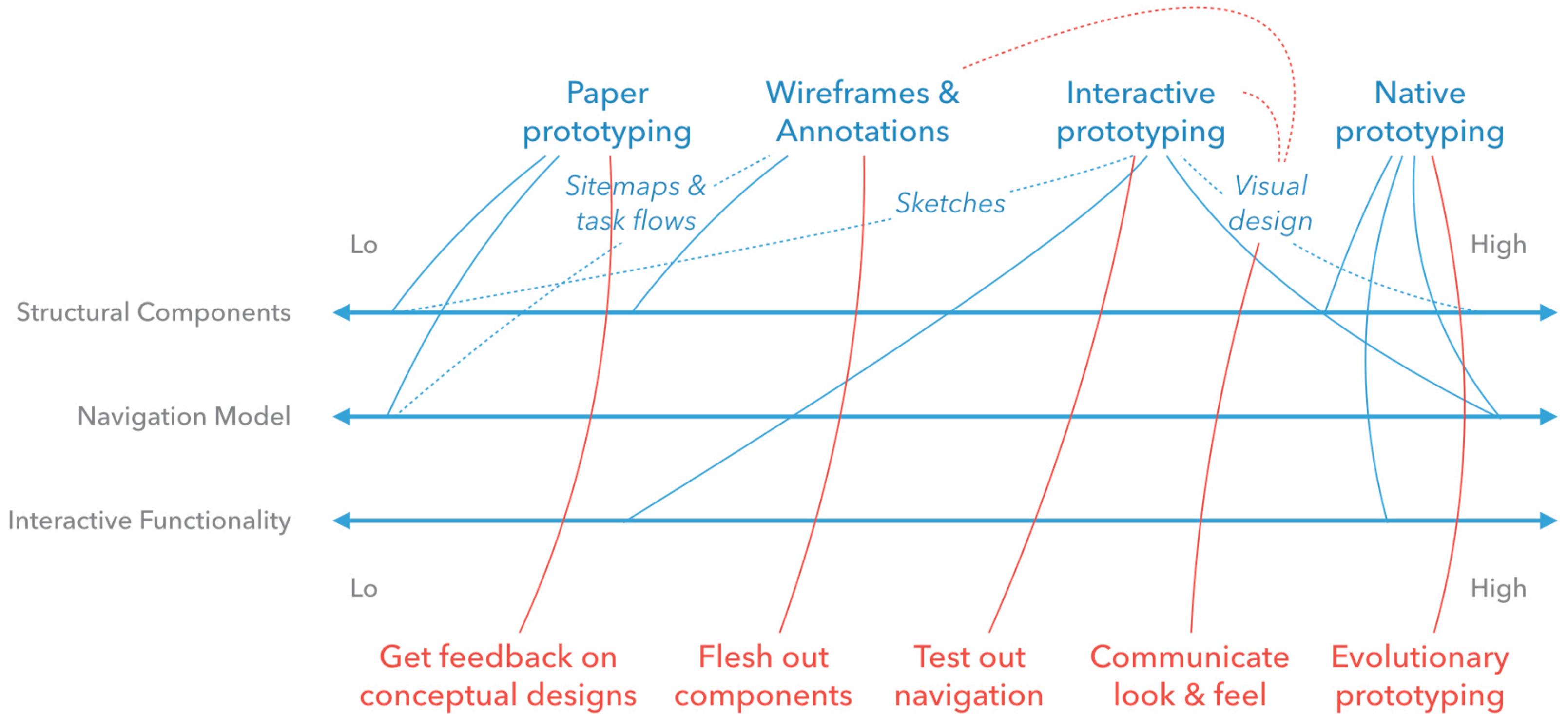
- Requires a facilitator to drive the prototype during testing and communication
- Offers limited ability to identify breakdowns in design
- Lacks sufficiently low-level specifications for development
- Provides limited sense of feasibility

Choosing the Right Method

How do we choose the right method?

- Various methods, approaches, and strategies for prototyping
- Criteria for choosing the right method:
 - Design team goals
 - Capabilities and resources
 - Available time for prototyping





ICA: Prototyping

Turn your *lo-fi* prototype into a *hi-fi* prototype!

Strategically use...

- Color
- Imagery
- Type

Questions?